

Prototype 4

Plan for 5

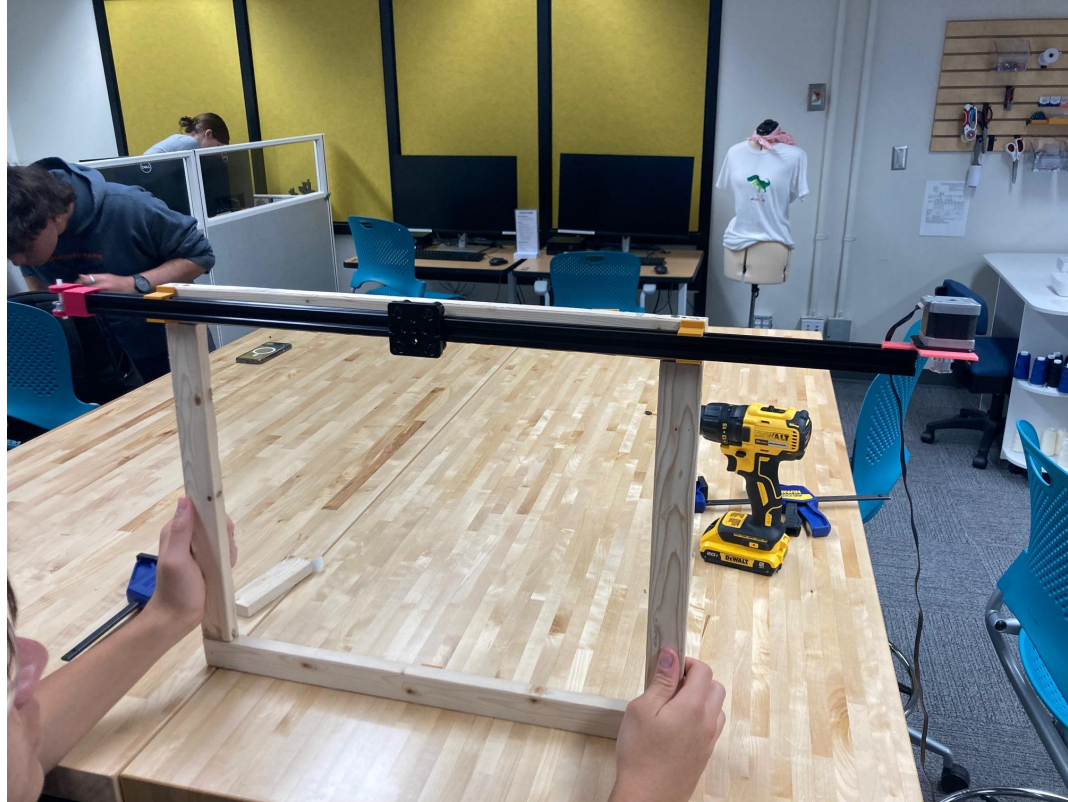
Palmeiras

Prototype 4

For Prototype 4, we:

- 3D printed two different prototypes for moving the squeegee along the x axis
 - One was to attach to another t-slot to the opposite end of the window
 - Another was to use a spring to drag the squeegee along without another t-slot
- Attempted to put together the piping system
- Built the window and frame, including fixturing
- Finalized code and attached limit switches

Window Built and Mounted!



Arduino Code Finalized with Limit switches!

- Limit switches to be glued onto T-slot, Exact location will be determined once we have final squeegee system in place.

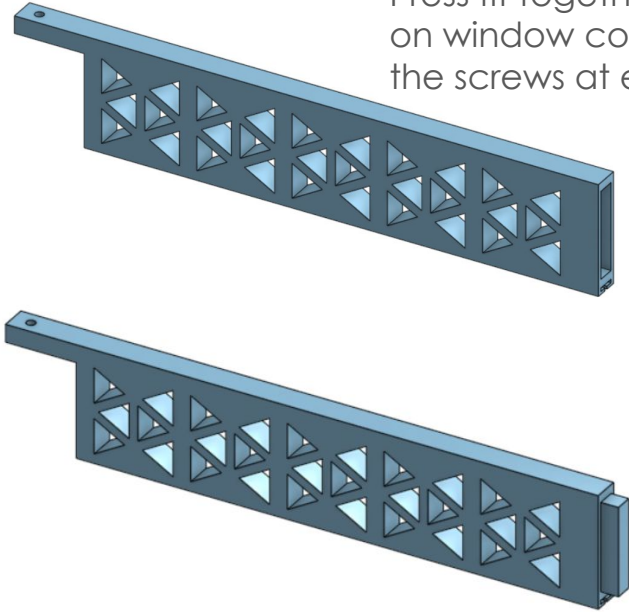


```
TwoSwitch.ino
1  const int stepPin = 2;
2  const int dirPin = 3;
3
4  const int limitSwitch1Pin = 7; // One end
5  const int limitSwitch2Pin = 8; // Other end
6
7  bool direction = LOW; // arb start in a direction
8
9  void setup() {
10     pinMode(stepPin, OUTPUT);
11     pinMode(dirPin, OUTPUT);
12
13     pinMode(limitSwitch1Pin, INPUT_PULLUP); // N.O.
14     pinMode(limitSwitch2Pin, INPUT_PULLUP); // N.O.
15 }
16
17 void loop() {
18     digitalWrite(dirPin, direction);
19
20     while (true) {
21         // check if either limit switch is pressed
22         if ((digitalRead(limitSwitch1Pin) == LOW && direction == LOW) ||
23             (digitalRead(limitSwitch2Pin) == LOW && direction == HIGH)) {
24
25             // reverse direction
26             direction = !direction;
27             digitalWrite(dirPin, direction);
28
29             // unpress switch
30             for (int i = 0; i < 50; i++) { // step count
31                 singleStep();
32             }
33
34             delay(200); // pause
35             break;
36         }
37
38         singleStep(); // continue
39     }
40 }
41
42 void singleStep() {
43     digitalWrite(stepPin, HIGH);
44     delayMicroseconds(900);
45     digitalWrite(stepPin, LOW);
46     delayMicroseconds(900);
47 }
48
```

Prototype 3D Print Designs

“Bridge Style”

Press fit together, pressure on window controlled by the screws at each t slot



“Hinge Style”



Rotating arm pretensioned with a spring

Pitfalls of Prototype 4

- The piping system didn't arrive on time (ordered on Monday, hadn't arrived by Friday)
- 3D print for one of the designs didn't fit correctly, long prints were time limited out of makerspace - turnaround time much longer than short prints.

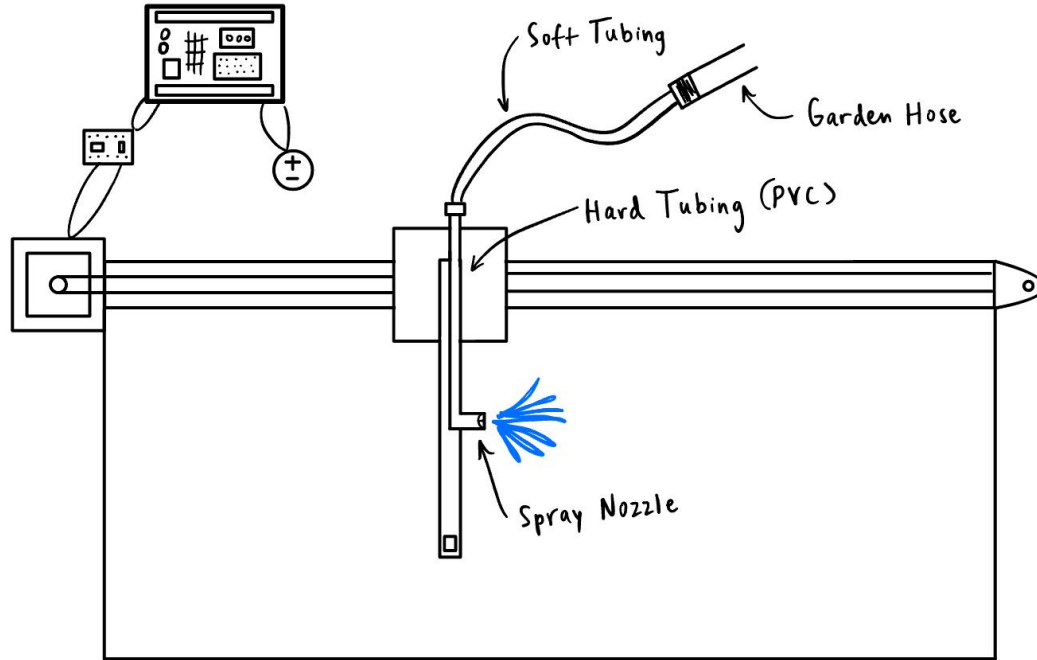
Fixing Shortcomings of Prototype 4

- Re-printing failed items
- Attaching tubing together

Prototype 5

- Picking one system to move the squeegee across the window (with or without second t-slot)
- Attach piping system to the window with either of the system we chose, adjust position and angle to get maximum coverage. Also test stationary position if spray is big enough.
- We have four different pieces (t-slot/gantry, window, tubing, and squeegee) which we will fit and attach together for the final prototype.
- Build housing for arduino, make sure it is spray proof.

Prototype 5 Sketch (The Final Prototype)



*Arduino Housing not pictured, it is a simple box

Timeline

Week 8

- End of Monday
 - Commit to which moving system to use
 - Figure out multiple angles to attach the tubing into either system
- Beginning of Wednesday Work Session
 - Have all 3D printed items ready
- End of the week
 - Attach the system together and make sure it all works
 - Iterate until it's polished by Wednesday